

Faculty of Computer Science and Engineering

Universitas Multi Data Palembang

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**IMPLEMENTATION OF AN AUTOMATED PLANT IRRIGATION
SYSTEM USING RANDOM FOREST**

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Abstract

Agriculture plays an important role in meeting food and industrial demands. However, plant irrigation that's still carried out manually causes inefficiencies in water usage and labor, leading to decrease productivity. The purpose of this research is to predict plant water requirements using Random Forest, which is implemented an ESP32 device. The testing stage compared max_depth 3 to 7 to determine the best parameters, while also considering computational efficiency. The results show that training accuracy at max_depth 3 to 5 reaches 99.96%, and at max_depth 6 to 7 it reaches 100%, accompanied by an increased AUC (Area Under Curve) value. However, max_depth values from 3 to 7 show the same testing accuracy, precision, recall, and f1-score of 99.83%, as well as consistent classification results in the confusion matrix. The research show that max_depth = 3 was chosen as the best parameter for maintaining generalization in predicting water needs for plants with low complexity, making it suitable for implementation on the ESP32 device.

Keywords: Agriculture, Water Requirement Prediction, Random Forest, Max Depth.